Why Did American Student Achievement Decline?

Verifying the Trends and Testing Competing Explanations with NAEP Data

June 2026

Abstract

American student achievement has fallen substantially over the past decade. Using data drawn directly from the National Assessment of Educational Progress (NAEP) Data Service API—national means and percentiles for mathematics and reading in grades 4 and 8 (1990–2024), the Long-Term Trend (LTT) assessment (1971–2023), state-level means, and demographic subgroups—this report first verifies the decline and characterizes its timing, distribution, and geography, then systematically evaluates eight candidate explanations. The data show a *two-phase* decline. Phase one (roughly 2013–2019) was a slow erosion concentrated almost entirely among the lowest-performing students: at age 13, scores at the 10th percentile of the LTT mathematics distribution fell 12.6 points between 2012 and 2020 while the 90th percentile was flat. Phase two (2020–2022) was an abrupt, across-the-board drop coinciding with pandemic schooling disruptions, followed by a weak and incomplete recovery that has stalled at the bottom of the distribution. By 2024, grade 8 reading stood at its lowest level ever recorded and the 90th–10th percentile gap in every grade–subject combination was the widest on record. Testing hypotheses against the timing, distributional, subgroup, geographic, and international evidence: pandemic disruption clearly caused phase two but cannot explain phase one; chronic absenteeism roughly doubled after 2020 and is a leading brake on recovery; demographic change, school funding cuts, and teacher shortages contribute little. Three further discriminating tests sharpen the verdict on phase one. A cohort-matched decomposition shows that pre-pandemic grade 8 losses arose *during* the middle-school years—cohorts arrived at grade 4 at record levels and lost ground between grades 4 and 8—identifying a period effect concentrated in adolescence. Catholic schools, never subject to federal test-based accountability, declined in step with public schools before the pandemic, and an event-study analysis of states' staggered NCLB-waiver adoption finds no effect of accountability relaxation on bottom-decile scores. A fourth test finds states that never adopted the Common Core declined as much as adopters, and the 2023 PIAAC shows the same bottom-concentrated decline among adults in nineteen OECD countries—a population untouched by any school policy. Together these results weaken every school-policy hypothesis and leave the post-2012 saturation of daily life by smartphones and digital media—which uniquely predicts an adolescent-led, sector-neutral, geographically uniform, international decline extending to adults—as the explanation most consistent with the full pattern of evidence, compounded by a pandemic shock whose effects have not been remediated. Two mechanism checks complete the picture: between-state variation in 2020–21 virtual schooling explains almost none of the between-state variation in NAEP declines (the closure effect, well identified at the district level, washes out under state aggregation), while NAEP's own attendance item shows student-reported absence rising from 2015–2019—before the pandemic—and a Kitagawa decomposition attributes 25–42% of the 2019–2024 declines, but only 7–19% of the pre-pandemic grade 8 declines, to the shift of students into high-absence categories.

1 Introduction

For two decades beginning in the early 1990s, American students made steady, historically large gains on the National Assessment of Educational Progress (NAEP), the nation's longest-running and most credible barometer of K–12 achievement. Fourth-grade mathematics scores rose nearly a full standard deviation between 1990 and 2013. That era is over. Every major NAEP series peaked between 2012 and 2015 and has fallen since, with the steepest losses after 2019. The reversal is large: by 2024, average grade 8 mathematics scores had returned to their 2000 level, and grade 8 reading scores were the lowest ever recorded in the assessment's history.

This report does two things. First, it *verifies* the decline directly from primary data, pulling national, state, percentile, and subgroup statistics from the NCES NAEP Data Service API rather than relying on secondary summaries (National Center for Education Statistics, 2025b). Second, it *tests* eight candidate explanations against five kinds of evidence: (i) the timing of the decline; (ii) its distribution across the achievement spectrum; (iii) its incidence across demographic subgroups; (iv) its geography across states; and (v) its presence or absence in other countries. Most public discussion attributes the decline to the COVID-19 pandemic. A central finding of this report is that the pandemic explanation, while clearly correct for the 2019–2022 drop, is incomplete: between a quarter and three-quarters of the total decline (depending on grade and subject) either predates March 2020 or postdates the return to in-person schooling, and the pre-pandemic phase has a distinctive signature—losses concentrated almost entirely among low performers—that any complete explanation must fit.

The analysis is descriptive and abductive rather than causal in the experimental sense: national test-score trends do not admit randomized variation, and several candidate causes operate simultaneously. The discipline imposed here is that each hypothesis generates predictions about timing, distribution, subgroups, geography, and international comparability, and is scored against all five. Hypotheses that fit one fact but contradict another are weighed accordingly.

2 Related Work and Contribution

The decline itself is well documented. NCES's own releases established the 2012–2013 peak and the concentration of losses among low performers; Malkus (2025b) assembles percentile trends across 21 assessments and distills four facts any explanation must fit (a ∼2013 onset, bottom-half concentration, internationally outsized U.S. gap growth, and parallel declines among *adults* on PIAAC—a fact Section 4.7 incorporates and sharpens with the 2023 Cycle 2 results); Wyckoff (2025) documents the same patterns at the state level; and the Education Recovery Scorecard project has traced the pandemic shock and recovery at district scale (Education Recovery Scorecard, 2024, 2025; Dewey et al., 2026). The hypothesis-weighing genre is likewise established—Malkus (2025b) and Wyckoff (2025) evaluate candidate explanations narratively against descriptive patterns—and the verdict this report reaches on the COVID and absenteeism hypotheses rests on existing causal work (Goldhaber et al., 2023; Jack et al., 2023; Council of Economic Advisers, 2023; Dee, 2024). The weak state-level correlation between pandemic schooling mode and NAEP changes was noted on release day in 2022 (by Chalkbeat's analysis and by the NCES commissioner; Barnum, 2022) and has an academic treatment in Kennedy et al. (2025); Section 8.1 confirms it with a pooled design and supplies the aggregation interpretation. On the digital-media hypothesis specifically, the nearest causal antecedents sit outside the U.S. achievement literature: Jain and Stemper (2024) estimate the effect of 3G arrival on PISA scores across

82 countries, and a 2025–26 rollout-based literature traces the same smartphone shock through fertility and family formation (Myers and Hooper, 2026; Hudson and Moscoso Boedo, 2026); Section 6.2 imports both. No within-U.S. mobile-rollout design against test scores previously existed—Wyckoff (2025) states the direct causal evidence is lacking—and Section 7.5 supplies a first pass: a county-level 4G-rollout event study built from the archived National Broadband Map, which returns a precise null on the arrival-timing margin while leaving the national and adoption margins open.

Three analyses in this report do not, to our knowledge, exist in published form, and each fills a gap that the existing literature has explicitly flagged:

First, the waiver-timing event study (Section 7.3). Dewey et al. (2026) make the descriptive observation that post-2013 declines were similar in waiver and non-waiver states, and write of the sustained-accountability counterfactual that it "has never been tested." The causal waiver literature is within-state—regression-discontinuity studies of priority/focus designations in Michigan, Kentucky, and Louisiana (Hemelt and Jacob, 2020; Dee and Dizon-Ross, 2019)—and Dee and Jacob (2011) identify NCLB's *adoption* effects from cross-state timing of prior accountability. Section 7.3 runs the Dee–Jacob design in reverse on the policy's dismantling, against state percentile scores, with randomization inference and an explicit power benchmark.

Second, the cohort/period decomposition (Section 7.1). Petrilli (2020) tracked two graduating classes informally and observed that their losses arose between grades 4 and 8; Malkus (2025b) argues against "cohort-cascade" accounts narratively. Section 7.1 provides the formal version: a cohort-matched decomposition of every grade 8 change into grade 4 entry and grades 4–8 growth, showing the declining cohorts arrived at record levels—the demonstration the narrative arguments lacked.

Third, the sector test (Section 7.2). Malkus (2015) noted after a single cycle (2013–2015) that combined public-plus-private estimates fell as much as public-only estimates and drew the same inference this report draws; Catholic-school NAEP commentary since has focused on COVID-era resilience. Section 7.2 executes the test systematically over the full pre-pandemic window, at the percentile level, with published-SE-based inference—navigating the fact that NAEP stopped reporting all-private results after 2013, which makes the Catholic series the only usable private benchmark.

Beyond the three tests, the report's integrative device—deriving distinct predictions from each hypothesis across timing, distribution, cohort structure, sector, geography, and international comparisons, and scoring all hypotheses against all facts—is, we believe, a more disciplined version of an argument that has so far been conducted one hypothesis at a time. The closest prior empirical work on waivers is Bleiberg (2020), whose dissertation estimates a difference-in-differences of waiver receipt (binary, by 2013) on mean student-level NAEP scores through 2013 and finds no average effect—a result that corroborates ours at the mean; his published waiver work concerns school-improvement designations within waiver states (Bleiberg, 2026). What remains distinct here is the staggered-timing-plus-never-treated design carried through 2019, the percentile outcomes that the bottom-concentrated decline makes essential, and the randomization-inference and power analysis.

3 Data

Main NAEP. The primary data are national public- and private-school average scale scores and percentile scores (10th, 25th, 50th, 75th, 90th) for mathematics and reading at grades 4 and 8, for all assessment years from 1990 (mathematics) and 1992 (reading) through 2024, retrieved from the NAEP Data Service API (National Center for Education Statistics, 2025b). Subgroup means (race/ethnicity, sex, National School Lunch Program eligibility), school-sector results (public, Catholic, private), state-level means for 2013–2024, and a state × year panel of percentile scores (2003–2024) were retrieved from the same source; state ESEA-waiver approval dates were compiled from Department of Education records, CRS Report R42328, and EdWeek's state-by-state tracking.[1] NAEP scale scores are reported on 0–500 scales; the within-grade student standard deviation (SD), estimated from the 2013 percentile spread, is roughly 29 points for grade 4 mathematics, 36 for grade 8 mathematics, and 34–36 for reading. A useful rule of thumb is that 10–12 NAEP points correspond to roughly one grade level of learning.

Long-Term Trend (LTT) NAEP. The LTT assessment has measured 9-, 13-, and 17-year-olds with substantially unchanged content since the early 1970s, providing the longest consistent yardstick. National means and percentiles for ages 9 and 13 (1971–2023) were compiled from the Digest of Education Statistics and the LTT Data Service and cross-validated against published highlights (National Center for Education Statistics, 2023). Two timing facts make the LTT unusually valuable here: the age 9 "2020" assessment was administered in January–March 2020, *immediately before* the pandemic school closures, and the age 13 "2020" assessment in fall 2019. These provide clean pre-pandemic endpoints.

International assessments. U.S. and OECD-average PISA scores (2000–2022) and U.S. TIMSS mathematics scores (2011–2023) are taken from NCES and OECD publications (OECD, 2023; National Center for Education Statistics, 2024b). The PISA 2022 public-use student micro-data file (613,744 students, 80 systems) is used directly for the device-distraction and screen-time analyses; estimates follow PISA conventions (ten plausible values combined by Rubin's rules, final student weights, school-clustered standard errors).

Contextual evidence. Published causal and descriptive research is used to test specific hypotheses: pandemic schooling-mode studies (Goldhaber et al., 2023; Jack et al., 2023), district-level recovery analyses (Education Recovery Scorecard, 2024, 2025), absenteeism tracking (Malkus, 2025a; Dee, 2024), accountability research (Dee and Jacob, 2011), and school-finance research (Jackson et al., 2016; Jackson and Mackevicius, 2024; Jackson et al., 2021).

[1] NSLP-eligibility breakdowns are available through 2022; the API does not return them for 2024. Pre-2003 national figures use the no-accommodations samples for 1990–1996 and accommodated samples thereafter, following NAEP reporting conventions. All scores were re-validated against published NAEP report-card values.

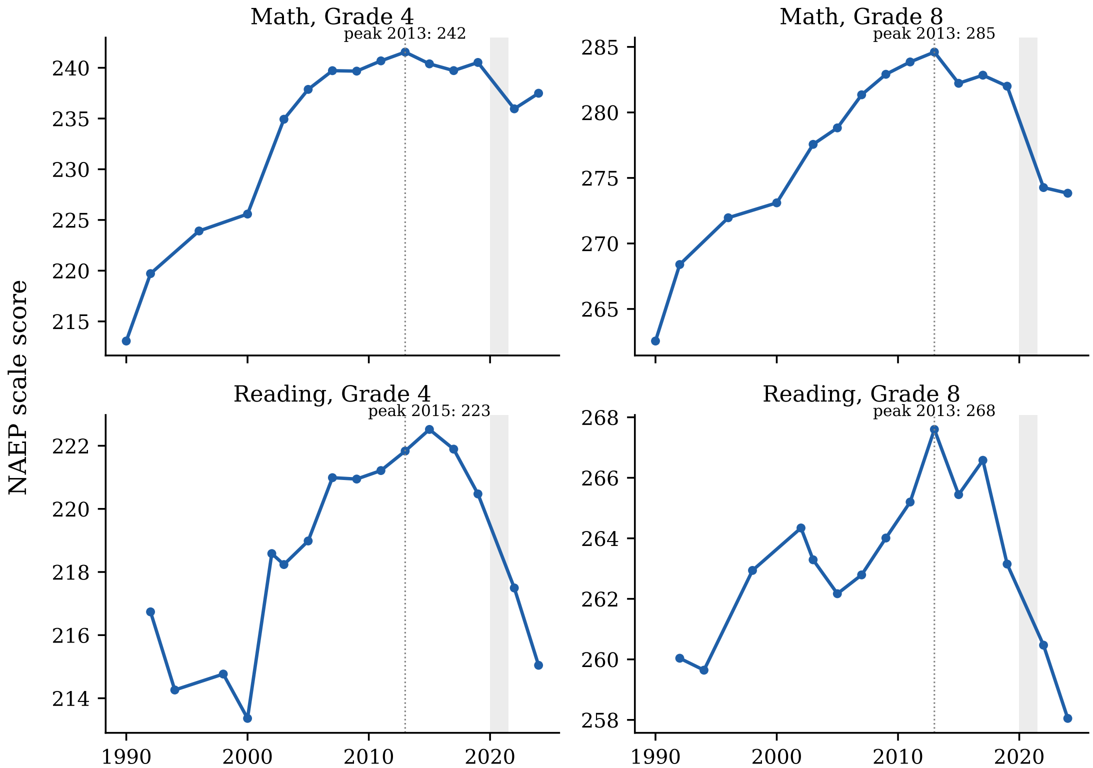

Figure 1: National average NAEP scale scores, 1990–2024. Dotted line marks 2013; shaded band marks the period of pandemic-disrupted schooling. Data: NAEP Data Service API.

4 Verifying the Decline

4.1 National trends: a peak around 2013, then two distinct drops

Figure 1 plots national average scale scores for the four main NAEP series. All four rose strongly through the 2000s, peaked in 2013 (grade 4 reading in 2015, fractionally above its 2013 level), drifted downward through 2019, dropped sharply between 2019 and 2022, and—except for a partial rebound in grade 4 mathematics—continued falling or stagnated through 2024.

Table 1 decomposes the decline. Three observations matter for everything that follows. First, the total losses are large: 0.14–0.30 student-level SD, roughly one grade level of learning in grade 8. In level terms, 2024 grade 4 mathematics matches 2005; grade 8 mathematics matches 2000;

Table 1: National average score changes by period (NAEP scale points).

	Change by period			Total 2013–2024	
Series	2013–19	2019–22	2022–24	Points	SD units
Mathematics, grade 4	−1.0	−4.6	+1.5	−4.1	−0.14
Mathematics, grade 8	−2.6	−7.7	−0.4	−10.8	−0.30
Reading, grade 4	−1.4	−3.0	−2.5	−6.8	−0.19
Reading, grade 8	−4.4	−2.7	−2.4	−9.5	−0.28

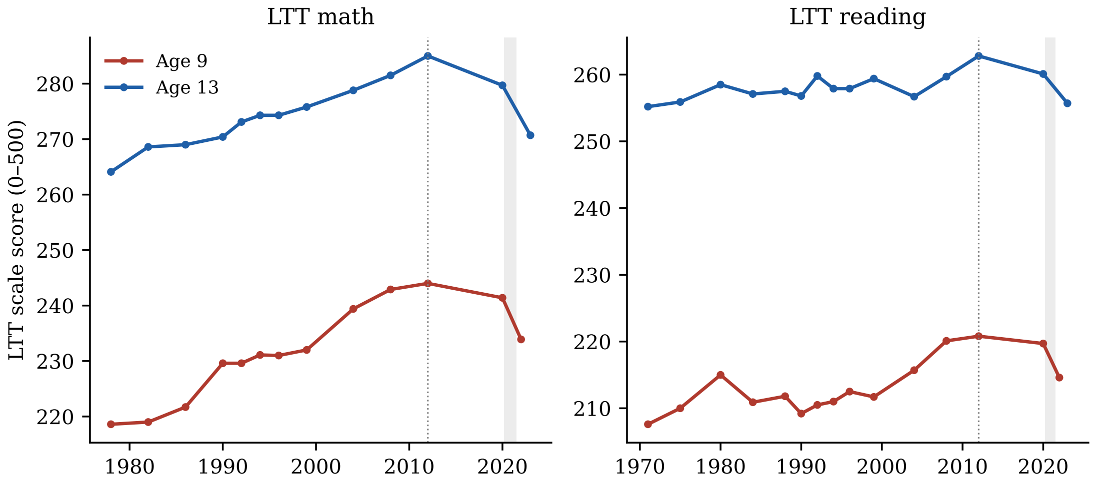

Figure 2: NAEP Long-Term Trend average scores, ages 9 and 13, 1971–2023. The age 9 "2020" point was assessed January–March 2020 (pre-pandemic); the age 13 "2020" point in fall 2019. Dotted line marks 2012. Data: NCES Digest tables 221.85/222.85, validated against the LTT Data Service.

grade 4 reading is below every year since 1998; and grade 8 reading (258.0) is below its previous historical minimum. Second, the pandemic window accounts for the majority of the mathematics decline (71% of the grade 8 total) but *less than half* of the reading decline at grade 8, where 46% of the loss had already occurred by 2019. Third, the post-pandemic period has produced essentially no aggregate recovery: only grade 4 mathematics regained ground (about a third of its pandemic loss), while both reading series fell *further* between 2022 and 2024.

4.2 The Long-Term Trend assessment confirms the timing

The LTT data (Figure 2) rule out the possibility that the main-NAEP decline is an artifact of framework or administration changes. On test content held essentially constant since the 1970s, age 9 and age 13 scores peaked in 2012 and were already falling before the pandemic: age 13 mathematics fell from 285 in 2012 to 280 in fall 2019, and age 13 reading from 263 to 260, *before any school closed*. The pandemic then produced the largest drops ever recorded in the series: age 9 mathematics fell 7 points between early 2020 and 2022 (its first statistically significant decline ever), and age 13 mathematics fell 9 further points by fall 2022, landing at its 1992 level. Age 13 reading in 2023 (256) was statistically indistinguishable from its 1971 baseline—a half-century of progress erased for the median young adolescent.

4.3 The distributional signature: collapse at the bottom

The single most diagnostic fact in the data is *where* in the achievement distribution the losses occurred. Figure 3 plots score changes since 2013 at the 10th through 90th percentiles for grade 8; Figure 4 shows the same comparison for the LTT, split into the pre-pandemic (2012→2020) and pandemic (2020→2022/23) windows.

Before the pandemic, the decline was almost exclusively a bottom-of-the-distribution phenomenon. Between 2012 and 2020, age 13 LTT mathematics fell 12.6 points at the 10th percentile, 7.4 at the 25th, 4.4 at the median—and *rose* 0.1 points at the 90th. Main NAEP shows

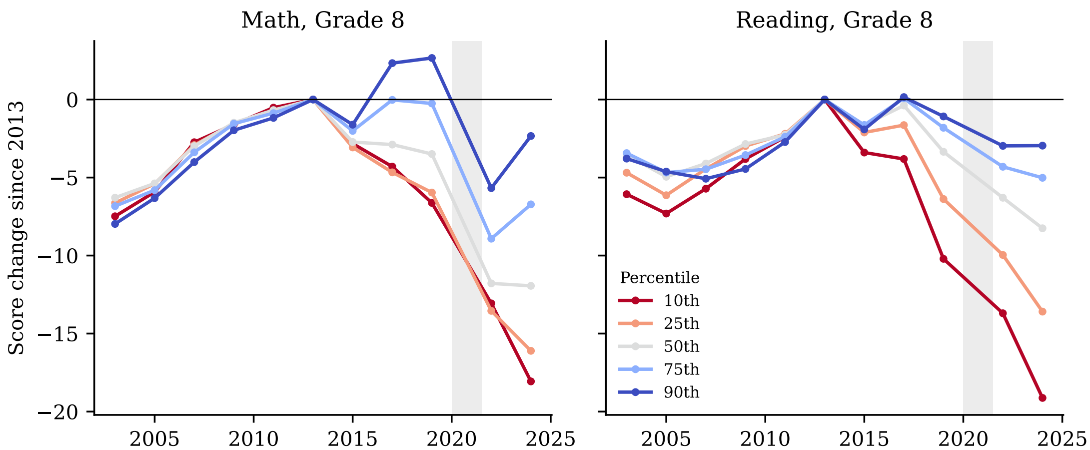

Figure 3: Grade 8 score changes since 2013 by percentile. Data: NAEP Data Service API.

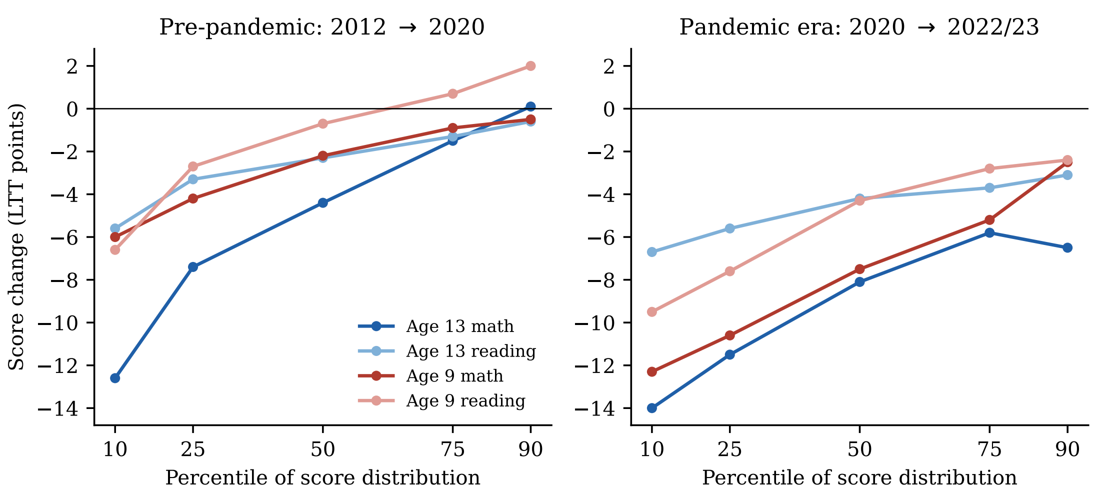

Figure 4: LTT score changes by percentile: pre-pandemic (left) versus pandemic era (right). The pre-pandemic decline is concentrated almost entirely among low performers; the pandemic-era decline hits everywhere but remains steepest at the bottom. Data: LTT Data Service.

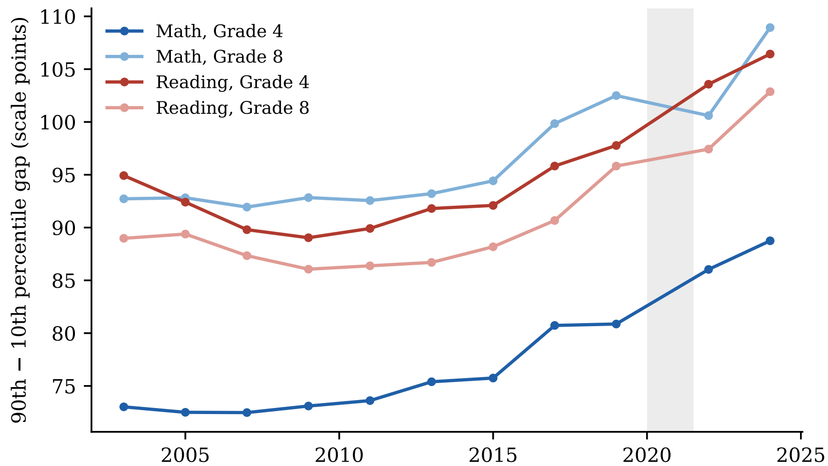

Figure 5: Gap between the 90th and 10th percentile scores, 2003–2024. Data: NAEP Data Service API.

the identical pattern: between 2013 and 2019, grade 8 mathematics fell 6.6 points at the 10th percentile while *gaining* 2.7 points at the 90th; grade 8 reading fell 10.2 points at the 10th percentile against 1.1 at the 90th. America's strongest students were holding steady or improving through 2019; its weakest students were in free fall years before anyone had heard of COVID-19.

The pandemic broke this pattern in an informative way: 2019–2022 losses appear across the entire distribution—in grade 8 mathematics the drop was essentially uniform, 6–9 points at every percentile, consistent with a shock (school disruption) that hit all students rather than only struggling ones. After 2022, the top of the distribution began recovering while the bottom kept falling: in grade 4 mathematics, 2022–2024 changes ranged from +2.7 points at the 75th percentile to −0.6 at the 10th.

The cumulative effect on inequality is unprecedented in NAEP's history. The 90–10 gap (Figure 5) widened between 2013 and 2024 by 13 points in grade 4 mathematics (75→89), 16 points in grade 8 mathematics (93→109), 15 points in grade 4 reading, and 16 points in grade 8 reading—in every case the widest gap ever recorded.

4.4 Subgroups: declines within every group

Score declines appear *within* essentially every demographic subgroup, which constrains compositional explanations (Section 6.5). Between 2013 and 2024, grade 8 mathematics fell 8.3 points among White students, 11.4 among Black students, 13.1 among Hispanic students, and was about flat (−1.1) only among Asian/Pacific Islander students. Both NSLP-eligible (lower-income) and non-eligible students lost ground through 2022 (grade 8 mathematics: −10.4 and −9.9 respectively). Pre-pandemic, the within-group pattern mirrors the percentile pattern: between 2013 and 2019 grade 8 mathematics scores among lunch-eligible students fell 3.6 points while non-eligible students lost only 1.0; Asian/Pacific Islander students *gained* 3.5 points. Figure 6 shows the level trends by lunch eligibility.

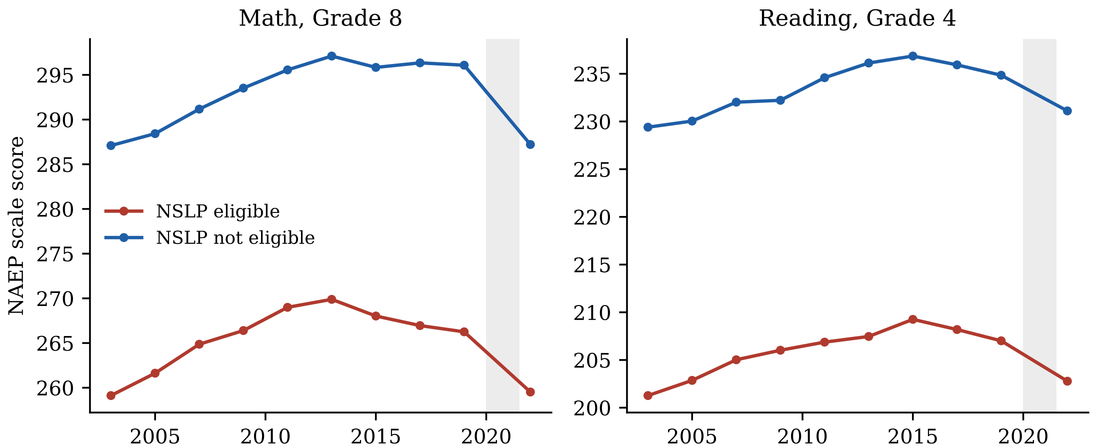

Figure 6: Average scores by National School Lunch Program eligibility (available through 2022). Data: NAEP Data Service API.

4.5 Geography: a national phenomenon, modulated by pandemic policy

The decline is not regional. Between 2013 and 2019, 75–94% of states (depending on series) posted declines; by 2024, between 88% and 100% of states remained below their 2013 level. No state escaped the national trend in grade 8 mathematics. The two prominent partial exceptions are instructive: Mississippi rose from 49th to 29th in grade 4 reading between 2013 and 2019 following its 2013 early-literacy reform (Spencer, 2024), and a handful of states (e.g., Alabama, Louisiana) regained or exceeded pre-pandemic grade 4 mathematics levels by 2024.

State variation *within* the pandemic window lines up with schooling-mode differences (Section 6.1). States' 2019–2022 drops in grade 4 mathematics ranged from about −13 points to roughly zero. The states that fell most recovered somewhat faster afterward (correlation between 2019–22 change and 2022–24 change ≈ -0.3 to -0.5; Figure 7), consistent with partial mean reversion as in-person schooling resumed—but the rebound slopes are shallow (about −0.4), implying that most of the pandemic-window loss has, so far, been persistent rather than transitory.

4.6 International context: America is not alone

The U.S. decline is part of a broader rich-world phenomenon (Figure 8). The OECD-average PISA mathematics score fell 22 points between 2012 and 2022 (494→472) and reading fell 20 points—declines that began *before* the pandemic. The OECD itself concluded that the fall was "only partly attributable to the COVID-19 pandemic," noting that reading and science had been sliding for roughly a decade (OECD, 2023). Twelve OECD systems—including Finland, the Netherlands, Belgium, and Canada—show statistically significant mathematics declines beginning before 2018. Notably, U.S. PISA *reading* was flat (505→504) between 2018 and 2022 while the OECD average fell 11 points, so the U.S. position relative to peers actually improved even as its NAEP reading scores fell; PISA samples 15-year-olds, whose NAEP-cohort losses were smaller than younger students'. U.S. TIMSS mathematics tells the same story as NAEP: grade 8 scores fell 27 points between 2019 and 2023, back to mid-1990s levels, with the lowest 10th-percentile scores since the study began (National Center for Education Statistics, 2024b).

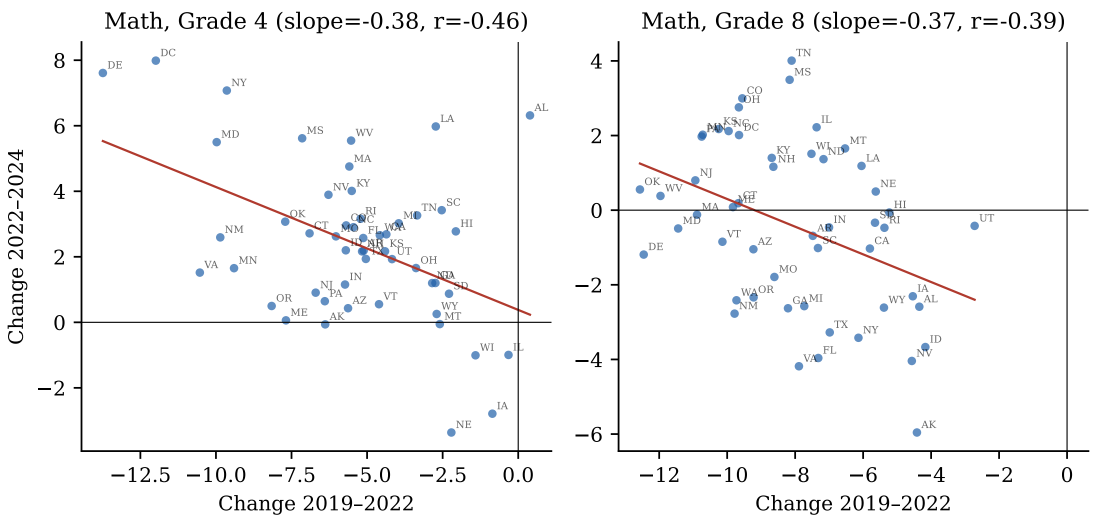

Figure 7: State-level pandemic losses versus post-pandemic recovery, mathematics. Each point is a state. Data: NAEP Data Service API.

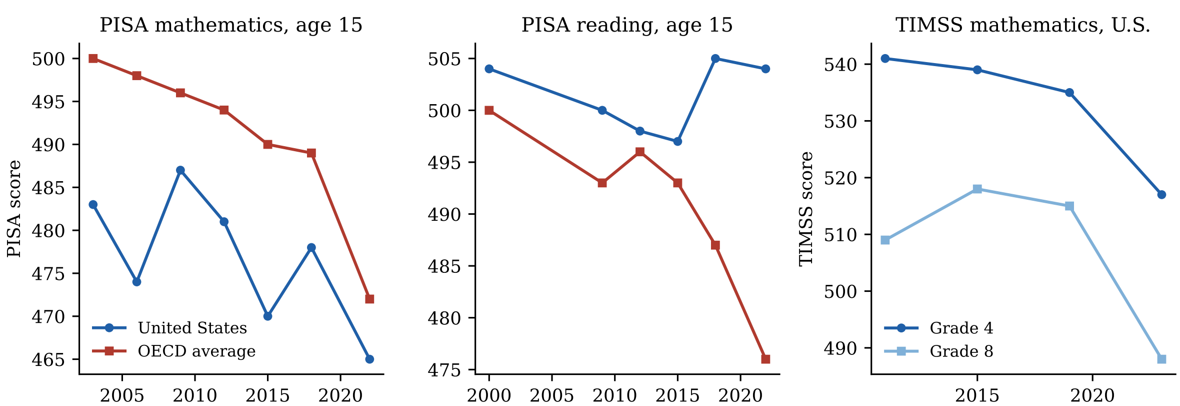

Figure 8: International assessments: PISA mathematics and reading (U.S. vs. OECD average) and U.S. TIMSS mathematics. Data: NCES/OECD publications.

4.7 Seniors and adults: the decline is not confined to grades 4 and 8

Two further populations complete the picture, and both carry unusual inferential weight because the usual school-policy explanations apply to them weakly or not at all. Grade 12 NAEP (2024 results released in September 2025) shows the same pattern as the younger grades: mathematics peaked in 2013 (153.5), drifted down through 2019 (150.3), and fell to its lowest level ever recorded in 2024 (146.9), with 2019–2024 losses of 5 points at the 10th percentile against no significant change at the 90th; reading fell to its own record low (282.6), and its 1992–2024 percentile changes (−24 at the 10th percentile, −2 at the 75th) are the three-decade version of the bottom-collapse (National Center for Education Statistics, 2025a).[2]

More striking is the adult evidence. The 2023 PIAAC assessment of adults aged 16–65 (National Center for Education Statistics, 2024a; OECD, 2024a) found U.S. literacy down 12 points from 2017 and numeracy down 6, with the decline concentrated almost entirely at the bottom—the share at or below Level 1 rose from 19% to 28% in literacy and 29% to 34% in numeracy while the share at Levels 4–5 was stable—and the same happened across the OECD: literacy fell in 19 OECD countries between 2012 and 2023, with the bottom decile declining and the top decile improving in most. Adults are not subject to school accountability, the Common Core, school funding, teacher quality, or school absenteeism. A force that simultaneously lowers the floor of measured literacy and numeracy among 40-year-olds in nineteen countries and among 13-year-olds in American public *and* Catholic schools is, almost by construction, not an education-policy variable. Malkus (2025b) flagged the adult parallel using earlier PIAAC rounds as one of his four keys; the 2023 cycle sharpens it considerably.[3]

4.8 Summary of facts to be explained

Any successful explanation must account for: (F1) a peak in 2012–2013 and slow decline through 2019; (F2) the pre-2019 losses concentrated overwhelmingly at the bottom of the distribution, with the top flat or rising; (F3) a large, across-the-distribution drop in 2019–2022; (F4) continued post-2022 decline in reading and at the bottom of every distribution, against partial recovery at the top and in younger students' mathematics; (F5) declines within every demographic group and nearly every state; (F6) similar pre-pandemic declines across many rich countries; (F7) record-high achievement inequality; and (F8) the same bottom-concentrated decline among high-school seniors and—in PIAAC—among adults across the OECD, populations to which most school-policy explanations do not apply.

5 Candidate Explanations

Eight hypotheses recur in research and public debate:

H1. **Pandemic disruption.** School closures, remote instruction, and associated social disruption caused learning loss.

[2] Grade 12 weighted student participation was 68% in 2024, well below the grades 4/8 rates; G12 levels warrant more caution than the younger-grade series.

[3] NCES cautions that cross-cycle PIAAC comparisons involve assessment and scoring changes; the trend-comparable estimates show the same pattern with a smaller literacy decline (9 points).

H2. **Smartphones and digital media.** The post-2012 saturation of adolescent life by smartphones and social media displaced reading, homework, and attention.

H3. **Accountability retreat.** The dismantling of No Child Left Behind's test-based accountability (waivers from 2012, ESSA from 2015) removed pressure that had propped up achievement among low performers.

H4. **Great Recession funding cuts.** State school-funding cuts after 2008 reduced school quality for the cohorts tested in the mid-2010s.

H5. **Demographic change.** A rising share of disadvantaged students mechanically lowered average scores.

H6. **Chronic absenteeism.** Students stopped attending school regularly after the pandemic.

H7. **Declining reading practice.** Children read less, for school and pleasure, eroding literacy skills.

H8. **Other: teacher shortages, lowered expectations/grade inflation.**

6 Testing the Hypotheses

6.1 H1: Pandemic disruption — decisive for 2019–2022, irrelevant before, incomplete after

Predictions. A sharp drop between the 2019 and 2022 assessments; larger losses where schooling was remote longer; recovery as schooling normalized; no effect before 2020.

Evidence. The 2019–2022 drop is unambiguous and its causal link to schooling disruption is among the best-documented findings in modern education research. Districts that were remote for more than half of 2020–21 saw mathematics achievement growth shortfalls of 0.44 SD in high-poverty schools, versus about 0.17 SD where instruction stayed in person (Goldhaber et al., 2023). Across twelve states, district math pass rates fell 14.2 percentage points on average in spring 2021, but only 4.1 points in districts that remained fully in person (Jack et al., 2023). The dose-response gradient, its replication across data sets, and the coincidence in time make H1 conclusively established for phase two—particularly for mathematics, which is learned mostly in school. Consistent with this, the 2019–2022 NAEP drop was nearly uniform across the achievement distribution (Section 4.3), as expected from a shock to schooling itself.

Two equally important negative results bound the hypothesis. First, H1 predicts nothing before 2020, yet roughly a quarter to a half of the total 2013–2024 decline predates the pandemic (Table 1), including the entire 2012–2020 collapse at the bottom of the LTT distribution. Second, H1 predicts recovery, and recovery has been feeble: by spring 2023 students had regained only about a third of pandemic mathematics losses and a quarter of reading losses (Education Recovery Scorecard, 2024); by 2025 only 17% of students were in districts at or above 2019 mathematics levels (Education Recovery Scorecard, 2025); and NAEP reading fell *further* after 2022. The ~ -0.4 slope in Figure 7 says the same thing in the NAEP data: states recovered only a fraction of what they uniquely lost.

Verdict. *Confirmed as the dominant cause of phase two (roughly half to three-quarters of the mathematics decline, less of reading); cannot explain phase one; the failure to recover requires additional explanation (H6, and the bottom-distribution forces of H2/H3).*

6.2 H2: Smartphones and digital media — the best fit to the international, pre-pandemic facts

Predictions. Decline beginning when smartphones saturated childhood (∼2012–2015); present across countries, subjects, and demographic groups; strongest in reading and among students with the least adult structure; displacement of reading and homework time.

Evidence on timing. Teen smartphone ownership in the U.S. went from 23% in 2011 to 37% in 2012, 73% by 2015, and 95% by 2018, with 45% of teens online "almost constantly" by 2018 (Madden et al., 2013; Anderson and Jiang, 2018). The achievement peak (2012–2013) coincides almost exactly with the takeoff. Adolescent entertainment screen time reached 7h22m/day by 2019 and 8h39m by 2021 (Rideout et al., 2022). The collapse in recreational reading is dramatic and—critically—happened mostly *before* the pandemic: the share of 13-year-olds reading for fun almost daily fell from 27% in 2012 to 17% in 2020 and 14% in 2023, while the share who never or hardly ever read rose from 22% to 31% (Figure 9).

Evidence on breadth. H2 is the only hypothesis that naturally explains fact F6, the synchronized pre-pandemic decline across rich countries with very different school policies, funding trends, and accountability regimes (OECD, 2023), and fact F8, its extension to adults. Within PISA 2022, 65% of students across the OECD (66% in the U.S.) reported digital-device distraction in mathematics class (OECD, 2024b). Re-estimating the score associations in the PISA 2022 student microdata (613,744 students; estimates combine all ten plausible values by Rubin's rules, use final student weights, cluster by school, and adjust for a quadratic in the PISA socioeconomic index): students reporting distraction in most or every mathematics lesson score 13.2 points lower in the U.S. (SE 3.9) and 15.0 points lower pooled across the OECD (SE 1.3) than less-distracted peers, and students reporting more than five hours of daily leisure device use at school score 40.5 points (SE 1.7) below light users—the OECD's published bivariate gaps survive socioeconomic adjustment essentially intact. These are associations, not causal estimates; the timing, cross-national synchrony, and adult parallel carry the causal weight.

Fit to the distributional signature. The microdata also speak directly to where the exposure sits. Within countries, every device-engagement measure is monotonically concentrated at the bottom of the score distribution: comparing students in their country's bottom vs. top mathematics quartile (OECD pooled, weighted), 32% vs. 24% report frequent classroom distraction, 8.2% vs. 2.7% report 5-plus hours of leisure device use at school, 24% vs. 12% report anxiety when their device is not nearby, and only 48% vs. 71% report turning off notifications in class. Critically, the same measures are nearly *flat across socioeconomic quartiles* (e.g., distraction 27.9% in the lowest ESCS quartile vs. 25.8% in the highest): heavy, poorly-regulated device engagement is a low-*performer* phenomenon, not a low-*income* phenomenon. This matches the decline's signature—bottom-concentrated within every demographic group (Section 4.4)—in a way no compositional story does. Causality plausibly runs both ways (struggling students may retreat to screens), and the broader literature finds displacement harms students with the least self-regulation most (Haidt, 2024); what the microdata establish is that the exposure profile and the damage profile coincide.

Causal evidence from phone bans. The hypothesis is no longer purely associational: a small quasi-experimental literature studies what happens when schools remove phones, and its results align with the decline's signature with striking precision. Staggered school-level bans in four English cities raised test scores by 6.4% of a standard deviation on average—and by *14.2% of a standard deviation for previously low-achieving students, with no effect on high achievers*

(Beland and Murphy, 2016): the mirror image of the bottom-concentrated decline. Staggered Norwegian middle-school bans raised girls' GPA (+0.08 SD), cut bullying by ∼0.4 SD, and reduced psychological-symptom GP visits, with the largest gains for low-SES girls (Abrahamsson, 2024); Spanish regional bans produced PISA gains worth more than half a year of learning (Beneito and Vicente-Chirivella, 2022); and the first U.S. causal evidence, from Florida's 2023 ban, finds test-score gains emerging in the second year alongside reduced unexcused absences—linking the phone and attendance channels directly (Figlio and Özek, 2025). The Swedish null (Kessel et al., 2020), in a setting where teachers could already confiscate phones, indicates the margin that matters is actually binding restrictions. The staggered state-level rollout of U.S. bans (one state in 2023–24, five by 2024–25, roughly twenty-five by 2025–26) against NAEP 2026 will provide a far larger test; the design and policy coding are specified in this project's repository in advance of the data.

Corroboration from an independent literature: fertility and family formation. A new quasi-experimental literature aims the same shock at an entirely different outcome and arrives at the same structure. Myers and Hooper (2026) exploit AT&T's exclusive iPhone carrier deal (2007–2011), under which early smartphone access varied with pre-existing AT&T coverage across counties: iPhone access reduced births by 4.5–8.0% at ages 15–19 and 3.2–6.6% at ages 20–24, with placebo tests on rival carriers' coverage returning nulls, and smartphone diffusion accounting for a third to a half of the total fertility-rate decline; the mechanism evidence points to reduced in-person interaction. Hudson and Moscoso Boedo (2026) instrument county 4G rollout with terrain ruggedness and find teen births fell first and fastest where high-speed mobile arrived earliest—in the United States and, in a parallel design, England and Wales—while the same instrument produces a rise in teen suicides, and time-diary data show teen in-person socializing roughly halving as digital leisure tripled. Three features of these results matter here. The *timing*: a break at 2007 compounding through the 2010–2015 4G wave, exactly the run-up to the 2013 achievement peak. The *age profile*: effects decline monotonically in age and vanish for adults—in the authors' words, whatever the smartphone shock is doing, "it is doing to teens"—the same adolescent-specific period effect that Section 7.1 finds in achievement. And the *identification*: rollout-based causal variation, not cross-sectional association. (The age profile is, notably, the chief *objection* to smartphones as an explanation of the adult-driven baby bust—and precisely the prediction the achievement hypothesis requires. The corroboration runs through the adolescent margin specifically: fixed-line broadband, a work tool for adults, if anything *raised* fertility among educated women (Billari et al., 2019).) Closest to the outcome studied here, Jain and Stemper (2024) link 3G arrival to 2.5 million PISA scores across 82 countries and estimate that mobile internet reduced mathematics, reading, and science scores by 0.04–0.08 standard deviations—about a quarter of a school year—concentrated among students with the least parental structure. Section 7.5 runs the first within-U.S. version of this design and finds a precise null on the local arrival-timing margin—a discipline on the hypothesis's mechanics (the harm did not ride on the cell-tower upgrade) rather than on its substance, since the national, adoption-driven channel that the evidence above concerns is invisible to that design by construction.

Verdict. *Strongly supported as a major driver of phase one (and of the continued post-2022 slide in reading): timing, international synchrony, mechanism evidence, quasi-experimental ban studies whose effect heterogeneity—low performers gain most when phones are removed—matches the distributional signature of the decline, and an independent rollout-based causal literature on fertility, mental health, and international test scores that recovers the same adolescent-*

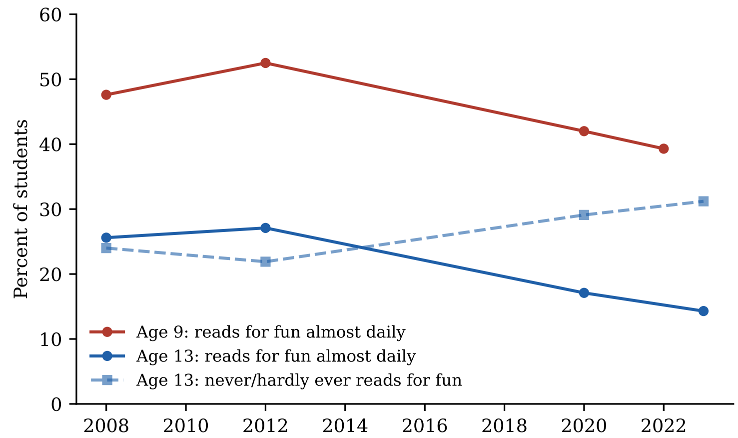

Figure 9: Share of students reading for fun, NAEP LTT student survey. The decline pre-dates the pandemic. Data: LTT Data Service (variable S003501).

specific, 2007-onset shock. The discriminating tests of Section 7—an adolescent-specific period effect, sector-neutral declines, and the failure of policy-variation tests—all land on the side this hypothesis predicts.

6.3 H3: Accountability retreat — the best fit to the bottom-concentrated U.S. signature

Predictions. Decline beginning after 2011–2012 (NCLB waivers) and deepening after 2015 (ESSA); concentrated among exactly the students accountability targeted—low performers and disadvantaged subgroups; U.S.-specific.

Evidence. The timing matches precisely: the first waivers releasing states from NCLB's consequences were granted in February 2012, covering 34 jurisdictions within a year; the Every Student Succeeds Act (December 2015) made the rollback permanent. The best causal evidence on NCLB found it had raised grade 4 mathematics by roughly 0.23 SD, with gains *concentrated among low-achieving, Black, Hispanic, and lunch-eligible students*, and no effect on grade 4 reading (Dee and Jacob, 2011). The post-2012 decline is nearly a mirror image of those gains: largest at the 10th percentile, among disadvantaged groups, and (initially) clearer in mathematics at the grades accountability touched most. If a policy raised the floor, removing it should drop the floor—and the floor is what dropped (F2). Commentators including Chad Aldeman and former IES director Mark Schneider have pressed this reading of the NAEP record (Aldeman, 2025).

The hypothesis has two limits. It cannot explain F6—Finland and the Netherlands did not pass ESSA—so it cannot be the sole driver. And because accountability operated through schools, it predicts little about the collapse in out-of-school reading (Figure 9). The Mississippi counter-example cuts in H3's favor: the one state that *intensified* early-grade stakes and instructional oversight after 2013 moved sharply against the national trend, with quasi-experimental evidence supporting a genuine policy effect (Spencer, 2024).

Verdict. *On the evidence of this section: supported as a contributor to the U.S. phase-one decline, especially its bottom-concentration; insufficient alone because the decline is international. Section 7 subjects this hypothesis to two direct tests—sector comparison and waiver-*

timing variation—and both come back unfavorable, substantially weakening it.

6.4 H4: Great Recession funding cuts — timing partially fits, magnitude does not

Predictions. Declines among cohorts schooled during 2009–2015 austerity, reversing as spending recovered after 2013.

Evidence. Real per-pupil spending bottomed in 2012–13 and 29 states cut per-pupil funding between 2008 and 2016 (Leachman et al., 2017); recession-induced cuts demonstrably reduced achievement (Jackson et al., 2021). But the magnitudes are far too small: the consensus causal estimate is roughly 0.03 SD per $1,000 per pupil sustained for four years (Jackson and Mackevicius, 2024). Recession-era cuts on the order of $1,000–$1,500 per pupil could account for perhaps 0.03–0.05 SD—a fraction of the 0.14–0.30 SD decline—and spending recovered steadily after 2013 while scores kept falling, the opposite of the predicted reversal. Funding also cannot explain F2's top-stability (cuts hit whole districts), F6, or the post-2020 dynamics, when an unprecedented $190 billion federal infusion accompanied historic losses (its measured effect, $\sim$0.008 SD per $1,000 per student, was positive but small relative to the shock).

Verdict. *Rejected as a major cause; plausibly a minor contributor to mid-2010s losses in the states that cut deepest.*

6.5 H5: Demographic change — arithmetic rules it out

Predictions. Stable or rising scores within each subgroup, with composition shifts toward lower-scoring groups driving the aggregate down.

Evidence. The within-group facts in Section 4.4 directly contradict the prediction: scores fell within every major racial/ethnic group, within both lunch-eligibility groups, and for both sexes. Composition shifts were also too slow (Hispanic share up five percentage points over the decade; English learners up about one) to move national means more than a point or two, and the direction of some shifts (record-low child poverty in 2021) runs opposite the score trend. A back-of-envelope reweighting of 2024 grade 8 mathematics subgroup means at 2013 race shares recovers less than 2 of the 10.8 lost points.

Verdict. *Rejected as an explanation of the decline (a 1–2 point compositional drag at most); within-group declines are the phenomenon itself.*

6.6 H6: Chronic absenteeism — a phase-two amplifier and the leading brake on recovery

Predictions. Score declines tracking the rise in absenteeism; weakest recovery where absenteeism stays elevated; no role before 2020.

Evidence. Chronic absenteeism was flat at roughly 15% from 2015 to 2019—so it cannot explain phase one—then nearly doubled to about 28.5% in 2021–22, retreating only to 23.5% by 2023–24 (Malkus, 2025a). The persistence of elevated absence is exactly what F4 (no aggregate recovery) needs: the Council of Economic Advisers calculated that elevated absence could account for roughly 27% of the 2019–2022 grade 4 mathematics decline and 45% of the reading decline (Council of Economic Advisers, 2023), and Dee (2024) and the Education Recovery

Scorecard identify it as a first-order obstacle to recovery. Absence is concentrated among low-income students, fitting the post-2022 divergence between the recovering top and the still-falling bottom.

Verdict. *Confirmed as a significant amplifier of phase two and probably the single most important proximate explanation for the failure to recover; explains little of phase one, though NAEP's own attendance item (Section 8.2) shows disengagement beginning to rise by 2017–2019, earlier than administrative data indicate. Absenteeism is partly a mediator—disengagement itself has causes (H1's disruption of attendance norms, H2's pull of screens).*

6.7 H7: Declining reading practice — well-documented, overlapping H2

The collapse in voluntary reading (Figure 9) is among the largest behavioral shifts measured in any NAEP survey: daily reading for fun at age 13 halved between 2012 and 2023, with most of the fall pre-pandemic. Reading achievement requires volume of practice, and the grade 8 reading series—where losses began earliest (F1) and 46% of the decline predates COVID—is where practice effects should show first. As an explanation this is best understood not as a rival to H2 but as its principal *mechanism* for literacy: the proximate cause is less reading; the distal cause is what replaced it. *Verdict: supported as the mechanism for reading's early and continuing decline.*

6.8 H8: Other candidates

Teacher shortages and quality. Vacancies and underqualified staffing are real ($\geq$36,000 vacant positions and $\geq$163,000 filled by underqualified teachers (Nguyen et al., 2024)) but became acute mainly after 2020 and are concentrated in special education, STEM, and high-poverty schools; teacher-preparation enrollment fell through the 2010s. Plausibly a modest contributor to phase two and to bottom-distribution stagnation; the timing fits phase one poorly. *Verdict: minor contributor.*

Common Core transition. The standards' implementation (2014–15 in most states) coincides with the decline's onset, making them a recurring suspect, particularly in mathematics. The hypothesis has usable state variation—four states never adopted, three repealed in 2014, and Minnesota adopted only the reading standards—and Section 7.4 tests it directly: pooled across grades and subjects, never-adopter states declined just as much (and more at grade 8). *Verdict: rejected as a major cause by its own state-variation test.*

Lowered expectations and grade inflation. High-school GPAs rose from 3.17 (2010) to 3.36 (2021) while ACT scores fell (Sanchez and Moore, 2022), and course rigor and homework time show parallel softening. This evidence indicates that *signals* of learning became disconnected from learning, masking the decline from parents and muting corrective pressure—an enabling condition more than a first cause. *Verdict: contributing background condition; direction of causality unclear.*

7 Sharper Tests: Cohorts, Sectors, and Policy Variation

The verdicts above lean on timing and incidence. This section reports three tests designed to discriminate between the two leading phase-one hypotheses (H2 digital media vs. H3 accountability), each exploiting a dimension of the data the candidate explanations treat differently.

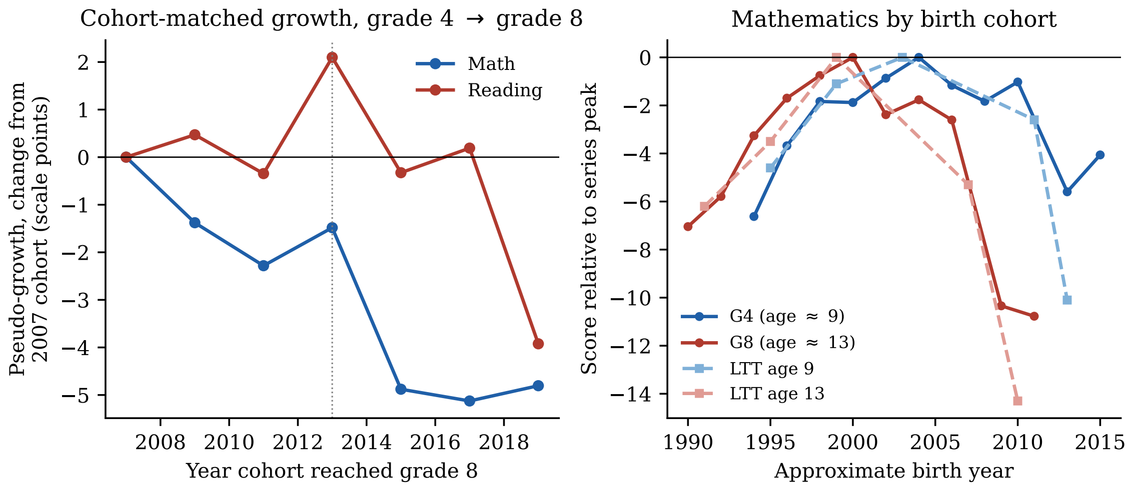

Figure 10: Cohort vs. period. Left: cohort-matched pseudo-growth (grade 8 mean minus the same cohort's grade 4 mean four years earlier), relative to the 2007 value. Right: mathematics scores by approximate birth year for four age groups, each relative to its series peak. Data: NAEP Data Service API; LTT.

7.1 Cohort vs. period: the losses accrue during adolescence

If successive birth cohorts were arriving at school weaker—because of early-childhood environment, poverty, or school quality in the early grades—declines should appear at grade 4 first and follow cohorts upward. If instead something in the post-2012 environment harms whoever is an adolescent at the time, grade 8 should fall regardless of how cohorts looked at grade 4. The data can separate these because the same birth cohort is observed at grade 4 and, four years later, at grade 8.

Define cohort-matched pseudo-growth as the grade 8 average in year t minus the grade 4 average in year $t-4$. Figure 10 (left) shows this quantity fell sharply for cohorts reaching grade 8 after 2013: mathematics pseudo-growth dropped from 44–46 points (2007–2013 cohorts) to about 41.5 points (2015–2019), and reading from 44–47 to 40.6 by 2019. Decomposing the 2013–2019 grade 8 decline: in mathematics, $-2.6 = +0.7$ (higher grade 4 entry scores of the later cohorts) -3.3 (less growth between grades 4 and 8); in reading, $-4.4 = +1.6 - 6.0$. *The cohorts that drove the pre-pandemic grade 8 decline arrived at grade 4 at record levels and then lost ground during the middle-school years.* The right panel shows the same fact by birth year: the identical cohorts that look healthy in the age-9/grade-4 series collapse in the age-13/grade-8 series once their adolescence falls after roughly 2012.

This is a period effect concentrated in adolescence, not a cohort effect. It is unfavorable to explanations rooted in early childhood or school inputs broadly (which would depress grade 4 entry too), and it matches the exposure profile of smartphones and social media, which saturate ages 11–14 far more than ages 5–9. It is compatible with the accountability hypothesis only in the version where pressure mattered most in the middle grades. The result formalizes what Petrilli (2020) observed informally for two graduating classes and supplies the demonstration behind Malkus (2025b)'s narrative argument against cohort-cascade accounts.

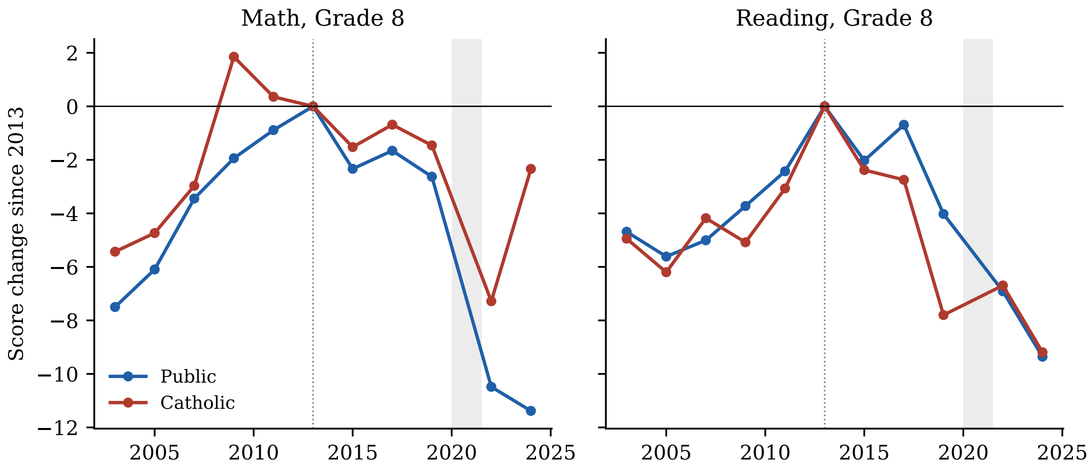

Figure 11: Average scores by school sector, relative to 2013. Data: NAEP Data Service API (SCHTYPE), national.

7.2 Public vs. Catholic schools: the decline is sector-neutral

Federal test-based accountability never applied to private schools, while the digital environment engulfed all adolescents. Malkus (2015) sketched this test informally after a single cycle; here it is run over the full window. NAEP reports Catholic-school results in every assessment year (overall private results are suppressed in several recent years for participation reasons), giving a usable untreated sector.[4]

Figure 11 shows changes since 2013, and because Catholic estimates carry standard errors near 1.6 points (vs. 0.3 for public; Digest table 222.32a), formal tests matter here. The statistically informative contrast is reading: Catholic grade 8 reading fell −7.8 points between 2013 and 2019 ($z = -3.4$ on published SEs)—at least as much as the public-school decline of −4.0—so the subject in which the pre-pandemic decline was largest and earliest was falling just as fast inside schools that NCLB and its repeal never touched. In mathematics the Catholic samples are too noisy to be informative: the −1.5 change is indistinguishable from zero and from the public −2.6. A pure accountability story predicts flat private-sector trends; the reading evidence contradicts that, while grade 4 (Catholic flat, public bottom decile falling) leaves room for an accountability contribution in the early grades. After 2019 the sectors diverge sharply and significantly in grade 8 mathematics: public schools fell 7.9 points more than Catholic schools ($z = -2.6$, using a conservative 2.5-point SE for recent Catholic estimates), by 2024 leaving Catholic students within 2.3 points of their 2013 level while public students were 11.4 points below—consistent with private schools' shorter closures and lower post-pandemic absenteeism.

7.3 NCLB waiver timing: no state-level accountability effect

The accountability hypothesis has a directly testable state-level implication: bottom-decile scores should begin falling when a state's accountability pressure was actually released. Dewey et al. (2026) observe descriptively that post-2013 declines look similar in waiver and non-waiver

[4] Catholic-school samples are far smaller than public ones, so their estimates are noisier, and Catholic enrollment is selected; level differences are uninformative. The comparison of *trends* is the test.

states and note that the sustained-accountability counterfactual "has never been tested"; this section provides the test. States received ESEA flexibility waivers at different times—eleven in early 2012, twenty-three more (plus DC) later in 2012, eight in 2013–2014—and seven states (CA, IA, MT, NE, ND, VT, WY) never received one and remained formally under NCLB until ESSA took effect in 2017–18.[5]

Figure 12 shows the test using the state-percentile panel (2003–2019, ending before the pandemic). Left: 10th-percentile grade 8 mathematics scores in early-waiver, late-waiver, and never-waiver states move essentially in lockstep—never-waiver states' bottom decile fell as much as waiver states' after 2012. Right: an event-study regression (early-waiver vs. never-waiver states, all four grade–subject cells pooled in within-cell SD units, state and year fixed effects, clustered by state) finds post-2012 coefficients near zero. A two-way fixed-effects estimate of the post-waiver effect on 10th-percentile scores is +0.09 SD with $p = 0.50$ pooled, and is small and insignificant in every cell separately; effects at the 25th and 90th percentiles are likewise null. The one prior quasi-experimental estimate—Bleiberg (2020), a binary waiver difference-in-differences on mean student-level NAEP scores through 2013—also found no average effect; the percentile results here extend that null to the part of the distribution where the decline actually occurred and through 2019.

Because the comparison group is only seven states, conventional cluster asymptotics are unreliable, so inference is re-done by randomization: permuting the waiver-timing assignment across states 2,000 times yields $p = 0.46$ for the pooled estimate ($p = 0.86$ for grade 8 mathematics in points). Crucially, the test is informative, not merely underpowered: the randomization distribution implies an 80%-power minimum detectable effect of about *3.2 points* on grade 8 mathematics 10th-percentile scores (0.35 SD pooled across cells)—less than half the ∼7-point gain Dee and Jacob (2011) attribute to NCLB's adoption, which was concentrated among exactly these students. If withdrawing accountability had undone even half of what imposing it achieved, this design would very likely have detected it. A joint test of pre-2011 event-study coefficients gives $p = 0.08$—early-waiver states were converging upward toward never-waiver states before treatment (visible in Figure 12)—a caveat that cuts both ways but does not rescue the hypothesis, since the post-2012 point estimates are wrong-signed for H3.

One scope limitation remains: a waiver is one notch of deregulation. If the operative change was a national shift in expectations and stakes culminating in ESSA, it hit all states simultaneously and is absorbed in year effects, invisible to this design. What the test rules out is the strong form of the hypothesis in which the formal release from NCLB consequences drove a state's low performers downward.

7.4 Common Core: states that never adopted it declined anyway

A third policy hypothesis with usable state variation is the Common Core: nearly all states adopted the standards in 2010–2011 and implemented aligned assessments around 2014–15, exactly when scores began falling, and critics have blamed the transition (particularly in mathematics) for the decline. Four states never adopted the standards (Alaska, Nebraska, Texas, Virginia), Minnesota adopted them in English language arts only, and three states formally

[5] Approval dates compiled from the Department of Education's per-state ESEA flexibility pages, CRS Report R42328, and EdWeek's state-by-state tracking. Washington's waiver was revoked in April 2014; it is dropped from the panel after 2013.

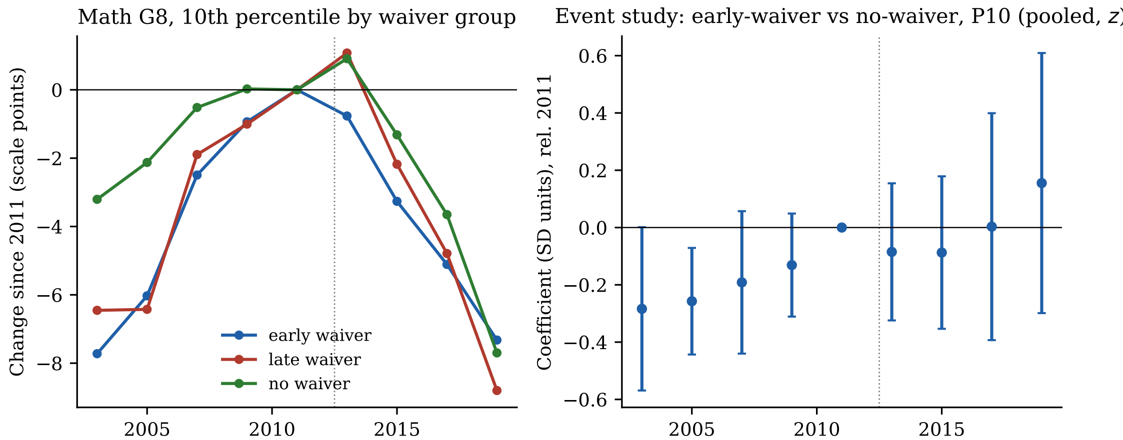

Figure 12: Waiver-timing tests, pre-COVID panel (2003–2019). Left: mean 10th-percentile grade 8 mathematics score by waiver group, relative to 2011. Right: event-study coefficients (early-waiver × year, base 2011, never-waiver comparison), all four grade–subject cells pooled in SD units; bars are 95% confidence intervals, clustered by state. Data: NAEP Data Service API; waiver dates from ED/CRS/EdWeek.

repealed them in 2014 (Indiana, Oklahoma, South Carolina).[6]

The test fails to implicate the standards (Figure 13). Between 2013 and 2019, never-adopter states' 10th-percentile scores fell less than adopters' in grade 4 mathematics, about as much in grade 4 reading, and *more* than adopters' at grade 8 (mathematics: −9.1 vs. −7.3; reading: −12.9 vs. −9.9); pooled across the four series, the never-adopter difference in bottom-decile change is −0.4 points ($p = 0.72$). The 2014 repealers look marginally better than adopters, but with three states the difference is uninformative. Minnesota's within-state subject contrast—its mathematics standards were never Common Core while its reading standards were—shows its mathematics performance falling *less*, relative to other states, than its reading performance (a +2.3-point triple-difference at grade 8), the opposite of what Common-Core-harmed-math would predict for an untreated subject, though a single state cannot carry much weight. As with the waiver test, the spillover caveat applies (never-adopter states still bought nationally aligned textbooks and materials); what the test rules out is the strong claim that adopting or keeping the standards is what drove a state's decline.

7.5 4G rollout timing: a first within-U.S. test of the digital-media channel

The fertility literature's identification strategy can be pointed at test scores. Using the archived National Broadband Map (block × provider mobile coverage, nine semiannual waves spanning the entire 2010–2014 LTE rollout, validated against FCC benchmarks, Verizon's December 2010 launch markets, and the later Form 477 data), we construct each county's high-speed-mobile coverage history and define treatment as the first wave in which coverage passes half the county's population. Matched to SEDA 5.0 county means (grades 3–8, mathematics and reading, 2009–2019; 99.9% of counties merge), this is, to our knowledge, the first within-U.S. mobile-rollout design run against student achievement. The identifying variation is the rural lag: metro counties

[6] Adoption and repeal dates from NCES State Education Reforms Table 2.17, cross-checked against contemporaneous coverage. Post-2015 "rebrands" that retained aligned content (Georgia, New Jersey, Arizona, and others) are coded as adopters.

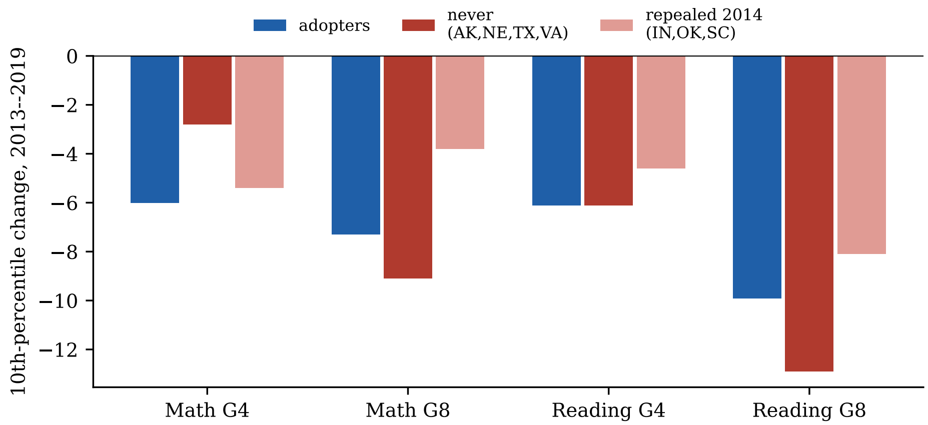

Figure 13: 10th-percentile score changes, 2013–2019, by Common Core adoption status. Data: NAEP Data Service API; adoption coding from NCES State Education Reforms.

typically crossed the threshold around 2011–2012, nonmetro counties one to three years later.

The result is a precise null on every margin the design can identify (Figure 14). The contemporaneous dose coefficient—moving a county from zero to full LTE coverage—is −0.002 SD (cluster SE 0.004; permutation $p = 0.66$), with an 80%-power minimum detectable effect of 0.013 SD; a within-county-year design identifying from cross-grade differences in adolescent exposure yields +0.011 SD per exposed year (permutation $p = 0.12$; MDE 0.016). A ten-specification robustness grid (alternative speed-tier codings, Verizon-only coverage, 25%/75% thresholds, dropping five states with known coverage-coding artifacts, unweighted) keeps every coefficient within ±0.003 SD, with signs that flip across specifications. The only nominally significant patterns—a small *positive* drift for early-treated counties, reading-only, growing with time since treatment—are continuations of a visible pre-trend, carry the wrong sign for the harm hypothesis, and die when the artifact states are dropped: the signature of mildly diverging urban–rural trends, the design's flagged first-order threat, not of a treatment effect.

Three limits keep this null from being a refutation of the hypothesis it tests. First, any national component of the smartphone effect—the channel most of Section 6.2's evidence concerns—is absorbed by year fixed effects; the design speaks only to whether *local 4G arrival timing* mattered. Second, smartphone adoption ran substantially on existing 3G networks (teen ownership was already 37% in 2012), so coverage timing is a noisy and partly anticipated instrument for exposure. Third, controls are exhausted by 2015 (the never-treated remainder is 117 deep-rural counties, half a percent of population), so slowly accumulating effects materializing four or more years after arrival are outside the design's reach. What the test does establish is informative on its own terms: the decline did not propagate through differences in when high-speed mobile infrastructure arrived—in instructive contrast to teen fertility, where the same variation produces large effects (Hudson and Moscoso Boedo, 2026), and to the cross-country 3G margin, where Jain and Stemper (2024) find achievement effects of 0.04–0.08 SD, several times our MDE on the within-U.S. 4G margin. Whatever digital media did to American adolescents' learning, it arrived with the device and its saturation of social life, not with the local cell tower upgrade.

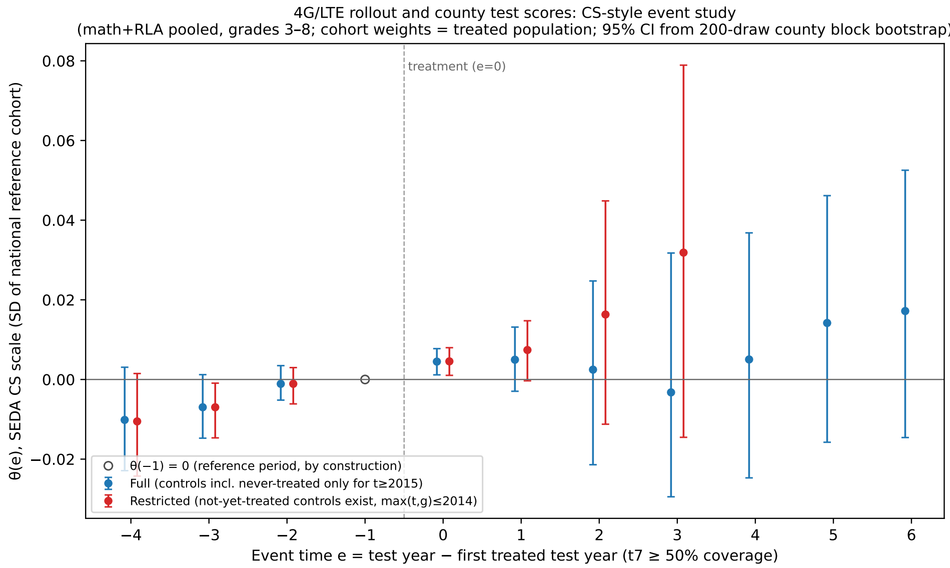

Figure 14: Event study of county 4G (LTE) arrival on SEDA county test scores, pooled grades 3–8, mathematics and reading, 2009–2019. Cohort-weighted ATT(g, t) aggregates by event time, with county-block-bootstrap 95% CIs; the restricted series uses only comparisons for which not-yet-treated controls exist. Treatment = first wave county high-speed mobile coverage exceeds 50% of population (NBM/SBDD archive). Units: SEDA cohort-standardized (SD) scale.

7.6 What the tests jointly imply

These tests were designed so the leading phase-one hypotheses predict different outcomes, and all of them landed on the same side. The pre-pandemic decline (i) strikes whoever is an adolescent after ∼2012 regardless of how the cohort looked at age 9, (ii) appears—in reading, the subject where the decline was earliest and largest—inside schools accountability policy never governed, (iii) shows no relationship to when states were actually released from NCLB, in a design powered to detect well under half the effect size accountability is credited with creating, and (iv) is equally present in states that never adopted the Common Core. Each fact is what the digital-media hypothesis predicts and what the school-policy hypotheses do not. The accountability retreat survives only in a weaker form—a national climate change contributing through channels common to all states, with its clearest remaining footprint at the public-school grade 4 bottom decile—while the case that the phase-one decline reflects a change in adolescent life outside the school building is now supported by timing, geography, sector, cohort structure, international synchrony, and the collapse of voluntary reading simultaneously.

8 Mechanism Checks: Schooling Mode and Attendance

Two further analyses probe the phase-two mechanisms directly: merging state NAEP changes with measured pandemic schooling mode, and exploiting NAEP's own student-reported attendance data.

8.1 Schooling mode: the district-level effect mostly washes out between states

Using the COVID-19 School Data Hub's district learning-model records aggregated to enrollment-weighted state shares (47 states plus DC; Iowa, Montana, and Oklahoma are uncovered), states' 2019–2022 NAEP changes can be regressed on the share of the 2020–21 school year spent fully virtual. The relationship is right-signed but weak in mathematics (-0.33 points per 10 percentage points virtual at grade 4, $p = 0.13$; -0.18 at grade 8, $p = 0.35$), absent in reading (grade 8 is wrong-signed), and essentially zero pooled across the four series (-0.04 points per 10pp, $p = 0.80$, clustered by state); Figure 16 (left) shows the grade 4 mathematics scatter. The result is robust to excluding DC and, if anything, strengthens under enrollment weighting (weighted slopes are zero in mathematics). Between-state variation in virtual share explains almost none of the between-state variation in score declines.

The same exercise can be pushed below the state level using NAEP's Trial Urban District Assessment (Figure 15): the 26 TUDA districts span essentially the full treatment range (Dallas, Miami-Dade, and Hillsborough were near-fully in person; Atlanta and Philadelphia fully virtual), and each is matched here to its own CSDH district record. Even with that contrast, the dose-response in NAEP is weak: -0.29 to -0.34 points per 10pp virtual at grade 4 ($p \approx 0.12$–0.16), zero at grade 8, -0.18 pooled ($p = 0.23$), and no relationship between virtual share and 2019–2024 net change. The confidence intervals are wide enough to admit effects of the size the growth-based literature reports, so the right reading is not contradiction but limitation: NAEP's cross-sectional district samples (with standard errors of 1.5–2.5 points) and the restricted, all-urban TUDA sample blur a gradient that longitudinal growth data resolve.

The basic state-level fact was observed when the 2022 scores were released (Barnum, 2022) and formalized by Kennedy et al. (2025); the pooled estimate here confirms it. It is not a refutation of the closure literature—the district-level studies (Goldhaber et al., 2023; Jack et al., 2023) exploit far larger within-state treatment variation with apter controls, and state aggregation compresses both the treatment (few states were mostly virtual on average) and the outcome (state NAEP changes carry sampling errors of roughly a point)—but it is a useful calibration: *schooling mode explains the timing of the national 2019–2022 drop better than it explains which states fell most.* Factors that varied little between states (the disruption itself, the absenteeism surge, whatever was already eroding the bottom) dominate the state cross-section.

8.2 Attendance: a pre-pandemic disengagement trend, visible inside NAEP

NAEP asks students how many days of school they missed in the month before the assessment, providing an attendance measure tied directly to scores and consistent since 2002. Three facts emerge (Figure 17). First, the share reporting three or more missed days, in the 19–22% range from 2003 through 2013 with no trend, *began rising before the pandemic*—to 24% at grade 4 by 2019—then jumped to 32–35% in 2022 and remained near 30% in 2024, mirroring the administrative chronic-absenteeism record (Malkus, 2025a) and revealing a disengagement pre-trend that the administrative data (flat at ∼15% through 2019) understate. Second, the score penalty associated with absence widened sharply *before* COVID: the gap between students missing no days and those missing more than ten grew from 29 to 42 points in grade 8 reading between 2013 and 2019—absence became more academically costly, or increasingly selected the most disengaged students, exactly when the bottom of the distribution was collapsing. Third, a Kitagawa decomposition splits each mean change into the part from students shifting into higher-absence categories and the part from score declines within categories: the absence shift accounts for roughly 25–42% of the 2019–2024 declines (e.g., 42% in grade 4 mathematics, 25% in grade 8 mathematics), closely bracketing the Council of Economic Advisers' estimate, but for only 7–19% of the pre-pandemic grade 8 declines—phase one was not an attendance phenomenon.

The decomposition is associational (absence proxies broader disengagement), but the pattern coheres with everything above: phase one operated through something other than attendance; phase two converted a disruption into a persistent attendance and engagement problem that the recovery has not closed.

8.3 Could it be test-taking effort rather than skill?

NAEP, PISA, and PIAAC are all low-stakes for the test-taker, and effort demonstrably matters for their *levels*: a surprise monetary incentive raised U.S. high-schoolers' scores on PISA-style items by roughly a quarter of a standard deviation (with no effect in Shanghai) (Gneezy et al., 2019); incentives raised grade 12 NAEP-style reading scores by at least 5 points (Braun et al., 2011); and effort proxies explain a third of cross-country PISA variation (Zamarro et al., 2019). If students simply began trying less hard on untested tests after 2012, part of the measured decline could be motivational rather than cognitive. This deserves a direct check rather than a footnote.

Three pieces of evidence bound the concern. First, PISA's "effort thermometer" asks students to rate (1–10) the effort they invested and the effort they would have invested had the test counted, in both 2018 and 2022. In the microdata, reported effort fell by just 0.13 points in the U.S. and 0.16 across the OECD; multiplying by the cross-sectional effort–score gradient (3.9 points per effort point, ESCS-adjusted—itself an upper bound on the causal effect) implies an effort-attributable change of roughly *half a point* of the 13–15-point 2018–2022 PISA decline, under five percent. Second, the OECD's behavioral indicators move the wrong way for an effort story: the within-test fatigue gap (first vs. second hour) *narrowed* in 2022—scores fell most at the *beginning* of the test—and answer straight-lining declined; the OECD concluded that administration conditions "remained similar," singling out only Albania (OECD, 2023). Rapid-guessing prevalence across countries is essentially uncorrelated with mean performance (Avvisati et al., 2024), and no NCES analysis documents rising disengaged responding on digital NAEP. Third, and decisively for the U.S. case: the same 2013–2019 decline appears in *state accountability tests*—the SEDA archive of roughly 70 million students' consequential state assessments (Dewey et al., 2026)—which carry stakes for schools, and among PIAAC adults, where the school-test motivation dynamic does not apply.

A caution survives: across countries, the change in even *hypothetical* ("if it counted") effort correlates 0.64 with the change in mathematics performance, so motivation and measured skill co-move—most plausibly because both are downstream of the same disengagement that the absence and reading-practice data record. The honest conclusion is that effort is a real wedge in score *levels*, but the change in effort is far too small to explain the decline; to the extent motivation has eroded, it is part of the phenomenon, not an artifact contaminating it.

8.4 Avenues examined and set aside

Several further hypotheses were tested or scoped and judged not to merit headline treatment; they are recorded here so the reader knows they were not overlooked. *Opioid epidemic / family*

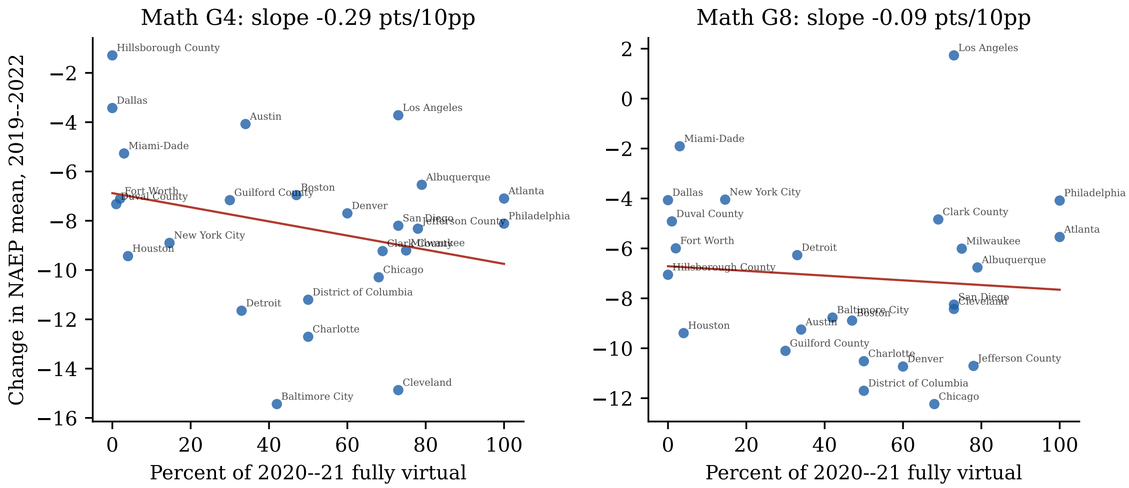

Figure 15: TUDA district dose-response: change in NAEP mathematics means, 2019–2022, against each district's CSDH share of 2020–21 spent fully virtual.

distress: regressing states' 2013–2019 bottom-decile changes on the change in CDC age-adjusted drug-overdose death rates yields right-signed but statistically null estimates in three of four series (the largest, grade 4 reading, has $r = -0.32$, $p = 0.12$); with heavy geographic confounding and no dose-response consistency, the test neither supports nor rules out a contributing role. *Cross-country smartphone-adoption timing vs. PISA declines*: country-level adoption series are too inconsistently measured across sources to support a credible event-time alignment; this remains the right design for future work with better data. *School discipline reform* (the 2014 federal guidance and suspension declines) has the right timing and a bottom-concentrated prediction, but no clean cross-state policy variation independent of the accountability and political variables already examined; it remains untested rather than rejected. *Cannabis legalization* fails on first inspection: adolescent use was roughly flat through the 2010s in national surveys, and legalization timing is collinear with state politics and pandemic schooling mode. *PISA smartphone-use frequency* (the ICT item available for U.S. students) was examined and discarded as an exposure measure: "uses a smartphone several times daily" is positively associated with scores (+70 points vs. never-users, SES-adjusted) because in a 95%-saturated population the abstainers are a highly atypical, disadvantaged group—frequency of use measures access, not excess; the hours-of-use and distraction items above do not share this defect.

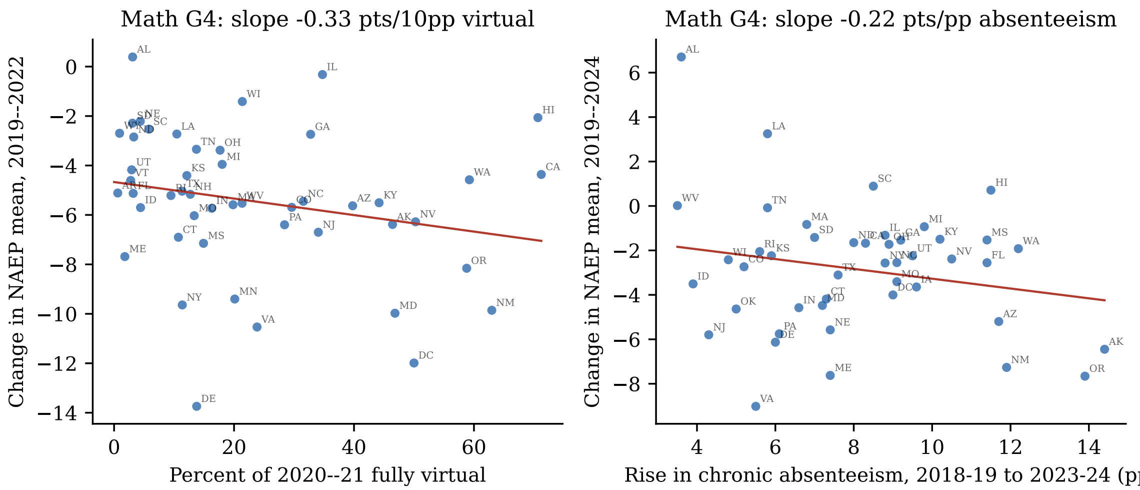

Figure 16: State dose-response. Left: change in grade 4 mathematics, 2019–2022, against the enrollment-weighted share of the 2020–21 school year spent fully virtual (COVID-19 School Data Hub). Right: net change 2019–2024 against the rise in chronic absenteeism, 2018-19 to 2023-24 (FutureEd state tracker).

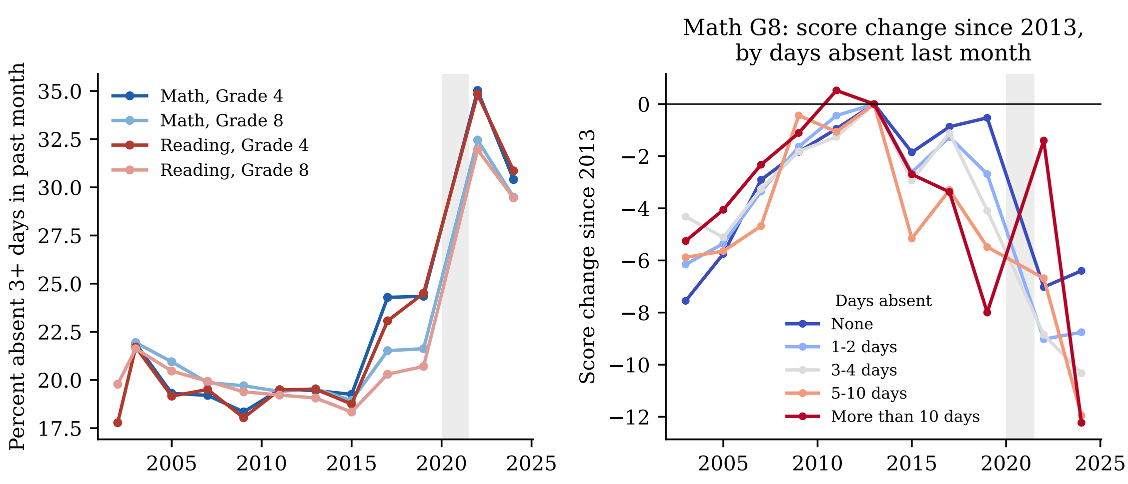

Figure 17: Student-reported absence inside NAEP (B018101). Left: share reporting 3+ missed days in the month before the assessment. Right: grade 8 mathematics score changes since 2013 by absence category. Data: NAEP Data Service API.

9 Synthesis: A Two-Phase Decline With Different Causes

A useful discipline can be borrowed from the parallel debate over collapsing birth rates, which shares this puzzle's structure—a synchronized post-2010 break across rich countries that resists every standard policy explanation (Kearney et al., 2022; Evans, 2024): a candidate cause must clear two bars *simultaneously.* It must explain the synchrony—the same decline appearing across countries with disparate school systems, funding trends, and accountability regimes, and among adults no school policy touches—and it must explain the local signature: bottom-of-distribution concentration, adolescent-period timing, sector neutrality. Every school-policy explanation fails the first bar by construction; the pandemic fails the second for phase one, having arrived six years late. Digital media is the only candidate examined here that clears both. With that framing, the evidence is not consistent with any single-cause account, but it is well organized by a two-phase model (Table 2):

Phase one (2013–2019): erosion at the bottom. Slow national decline, concentrated almost entirely among the lowest-performing students, present in most states and many countries, alongside collapsing voluntary reading. Two hypotheses fit the basic facts: the accountability retreat (which predicts the bottom-concentration and the U.S. policy timing, and is corroborated in mirror image by Mississippi) and digital-media displacement (which predicts the international synchrony and the reading-practice collapse). The discriminating tests of Section 7 break the tie in favor of digital media: the losses accrue during adolescence to cohorts that arrived at grade 4 at record levels, they appear equally inside Catholic schools that accountability never governed, they bear no relationship to when individual states were released from NCLB or to whether a state ever adopted the Common Core, and—the widest version of the same test—they reappear among adults across nineteen OECD countries (Section 4.7), a population no school policy touches. The accountability retreat retains a plausible supporting role—as a national climate shift, and at the public-school grade 4 bottom decile, where the sector contrast does point its way—but the strong version in which deregulation drove the floor down is rejected by its own state-level test. Funding, demographics, and teacher supply fail on timing, distribution, or magnitude.

Phase two (2019–2024): shock without recovery. The pandemic caused the large 2019–2022 drop—the schooling-mode dose-response evidence is decisive—and hit the whole distribution. What demands explanation is the aftermath: four-plus years on, aggregate recovery is negligible, reading is still falling, and the bottom decile continues to sink while the top re-

Table 2: Hypothesis scorecard against the facts of Section 4.

Hypothesis	F1 Timing	F2 Bottom-heavy	F3 COVID drop	F4 No recovery	F5/F7 Within-group	F6 Intern'l
H1 Pandemic	×	∼	✓	∼	✓	∼
H2 Digital media	✓	∼	×	✓	✓	✓
H3 Accountability	✓	✓	×	∼	✓	×
H4 Funding cuts	∼	×	×	×	∼	×
H5 Demographics	×	×	×	×	×	×
H6 Absenteeism	×	✓	✓	✓	✓	∼
H7 Reading practice	✓	∼	×	✓	✓	✓
H8 Teachers/inflation	∼	∼	×	∼	∼	×

✓ = consistent; ∼ = partially consistent; × = inconsistent or silent.

bounds. Persistently doubled chronic absenteeism is the leading proximate cause, with the phase-one forces (weak accountability pressure, screen displacement) plausibly explaining why disengagement has not self-corrected. On the magnitudes in Table 1, phase one accounts for roughly 20–45% of the total 2013–2024 decline depending on the series, the pandemic window for roughly 30–70%, and the post-2022 period for the remainder—meaning that even a complete reversal of pandemic losses would leave the United States well below its 2013 peak.

Three implications follow. First, "COVID recovery" framing understates the problem: grade 8 reading was already down 4.4 points before the pandemic, and the forces behind that slide are still operating. Second, because losses are concentrated at the bottom, average-score targets understate the equity emergency; the 90–10 gap is the widest ever measured. Third, with the policy-side explanations weakened by the tests of Section 7, the empirical frontier shifts to the digital-media mechanism, where causal evidence is accumulating: existing ban studies (Beland and Murphy, 2016; Abrahamsson, 2024; Beneito and Vicente-Chirivella, 2022; Figlio and Özek, 2025) consistently find gains concentrated among low achievers, and the staggered adoption of U.S. state bans (roughly thirty states by 2025–26) against NAEP 2026 offers exactly the kind of test that waiver timing provided here—this project's repository pre-specifies the design and treatment coding. Mississippi's literacy reform remains the strongest evidence that aggressive instructional policy can buck the trend even if policy retreat did not cause it.

10 Limitations

This analysis is observational. The hypothesis tests rest on timing, distributional incidence, and cross-sectional comparisons, not experimental variation; several causes plausibly interact (absenteeism is partly downstream of screens and of weakened school engagement), so the phase shares above are accounting decompositions, not causal attributions. NAEP percentile statistics for the oldest years use no-accommodation samples; the NSLP-eligibility series ends in 2022 and is affected by community-eligibility expansion; PISA OECD averages shift membership across cycles; and the LTT format changed in 2004 (results are bridged). Measurement artifacts can be largely ruled out: NAEP exclusion rates were flat at 1–3% of all students from 2013 through 2024 even as SD/EL identification rose, student participation was identical in 2022 and 2024 (92% at grade 4, 89% at grade 8), and public-school participation was complete—so neither rising exclusion nor falling participation can explain the declines (grade 12, with 68% student participation in 2024, is the exception and is treated cautiously). PIAAC cross-cycle comparisons involve assessment changes and should be read as NCES advises. The Section 7 tests carry their own caveats: the never-waiver comparison group is seven states—inference therefore uses randomization rather than cluster asymptotics, and the minimum-detectable-effect calculation bounds what the null can claim—with visible pre-2011 trend differences ($p = 0.08$ jointly); waivers capture only the formal step of deregulation; and Catholic-school samples are small and selected, so only their trends—not levels—are informative, and only the reading trends are estimated precisely enough to discriminate. The Section 8 state regressions are aggregation-limited (state-level treatment variance is a fraction of district-level variance, and three states lack schooling-mode data), NAEP's absence item covers only the month before the assessment, and the Kitagawa decomposition treats absence as exogenous when it partly proxies disengagement. Finally, the digital-media evidence combines associational gradients (the PISA distraction and screen-time analyses are cross-sectional; low performance may drive device retreat as well as the reverse, and the leisure-hours items were not administered to U.S.

students) with a small but consistent quasi-experimental ban literature and with rollout-based causal designs whose outcomes are either non-U.S. test scores or non-achievement domains (fertility, mental health); our own 4G-rollout test (Section 7.5) is informative only about local arrival timing, has imperfect pre-trends, and cannot see a nationally synchronized effect; the causal magnitudes at the population scale of the U.S. decline remain uncertain, and the PIAAC adult comparison carries its own caveats (a 28% U.S. response rate in 2023 and a change to tablet-only administration, with NCES advising caution on cross-cycle comparisons).

11 Conclusion

The decline in American student achievement is real, large, and verifiable from primary data: roughly one grade level of learning lost at grade 8 since 2013, the lowest grade 8 reading scores ever recorded, and the widest achievement gaps in NAEP's history. It happened in two phases with different causes. A pre-pandemic erosion, nearly invisible in averages because it was confined to the bottom of the distribution, is best explained by the saturation of adolescent life by smartphones and digital media after 2012: the losses strike whoever is an adolescent after that date (not particular cohorts), appear identically in Catholic schools that accountability policy never governed, show no relationship to when states were released from NCLB, recur across the rich world, and coincide with a halving of voluntary reading. The retreat from test-based accountability likely contributed at the margins—most visibly at the public-school grade 4 bottom decile—but failed the direct tests this report could put to it. The pandemic then imposed the largest schooling shock in modern history, whose effects, far from fading, have been locked in by chronic absenteeism that remains half again its pre-pandemic level. Funding cuts, demographic change, and teacher shortages are, on the evidence, second-order. The policy conclusion is uncomfortable but clear: returning to 2019 practices would only return the country to a trajectory that was already pointing down.

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